



# Carp-inspired self-regulating marine antifouling coating: Featuring robust controlled release of eugenol and high-efficiency self-healing performance

Jiahuan Liu<sup>a,b</sup>, Qiang Li<sup>a,b</sup>, Taijiang Gui<sup>c</sup>, Hongyu Guo<sup>d</sup>, Kai Zhang<sup>c</sup>, Fandong Meng<sup>a,b</sup>, Xiaoli Zhan<sup>a,b</sup>, Quan Liu<sup>a,b,\*</sup>, Qinghua Zhang<sup>a,b,\*</sup>

<sup>a</sup> Institute of Zhejiang University-Quzhou, Zhejiang Provincial Innovation Center of Advanced Chemicals Technology, Quzhou 324000, China

<sup>b</sup> College of Chemical and Biological Engineering, Zhejiang University, Zhejiang Provincial Key Laboratory of Advanced Chemical Engineering Manufacture Technology, Hangzhou 310027, China

<sup>c</sup> State Key Laboratory of Marine Coatings, Marine Chemical Research Institute Co. Ltd., Qingdao, China

<sup>d</sup> Xichang Satellite Launch Center, Xichang 615000, China

## ARTICLE INFO

### Keywords:

Marine antifouling  
Robust controlled releasing  
Self-healing  
Fluorinated eugenol

## ABSTRACT

The practical application of low surface energy polydimethylsiloxane (PDMS) in marine antifouling is significantly limited due to its inadequate static antifouling capability. Herein, integrating a strategy inspired by carps to release antifouling agents with the fouling release properties of low surface energy materials, an innovative self-regulating marine antifouling coating (named PDMS-Pun-x) was developed, which demonstrates remarkable robust ability of controlled release of eugenol and efficient self-healing performance. Under seawater, the coating regulates its inherent hydrophilicity and the content of the surface anti-fouling agent to obtain the ability of controlled releasing eugenol. Compared with other self-renewal coatings, there proves non-heavy metal leaching and no internal self-hydrolysis existing. Moreover, based on the synergistic action of disulfide bonds and hydrogen bonds, the coating demonstrated a notable self-healing performance, with a self-healing rate of 94.06%. Additionally, the substrate of PDMS modified with polyurea provides the coating with excellent resistance against corrosion. This study introduces a highly promising approach towards the advancement of environmentally-friendly and durable marine antifouling coatings.

## 1. Introduction

The formation of biofouling by bacteria, proteins, barnacles, and other marine organisms has been found to exert detrimental effects on marine industries and marine bioengineering. This phenomenon leads to an increase in surface roughness and hull mass, resulting in a substantial rise in operational costs and a reduction in the vessel's service life [1,2]. Owing to the significant pollution caused by conventional heavy metal antifouling agents like cuprous oxide and TBT, extensive research has been conducted on eco-friendly ship antifouling coatings [3–6].

Polydimethylsiloxane (PDMS) coating is widely considered as one of the most promising environmentally friendly antifouling coatings due to its notable characteristics including low surface energy, low Young's modulus, and cost-effectiveness [7,8]. However, in complex marine environments, the hydrophobic nature of PDMS makes it susceptible to contamination by slime-containing bacteria, proteins, and extracellular

polymeric substances. This slimy residue significantly accelerates the formation of biofilms and attracts the adhesion of larger marine fouling organisms such as barnacles and mussels [9]. In recent years, various natural antifouling agents, such as coffee, capsaicin, eugenol, and isoborneol, have demonstrated their effectiveness in enhancing the antibacterial and antifouling capabilities of marine coatings [10–12]. However, the conventional methods of physically mixing or covalently grafting these agents into coatings, though improving their antibacterial and antifouling performance to some extent, have raised concerns regarding the non-releasability and uncontrolled release rate of the antifouling agents, resulting in decreased antifouling efficacy and unnecessary wastage of the agents. The research and development of surface self-renewal coatings which offer the advantage of steadily releasing the antifouling agents have garnered significant attention [13–15]. While it is worth noting that most existing surface self-renewal coatings still heavily rely on metal ions and possess the problem of

\* Corresponding authors at: Institute of Zhejiang University-Quzhou, Zhejiang Provincial Innovation Center of Advanced Chemicals Technology, Quzhou 324000, China.

E-mail address: [qhzhang@zju.edu.cn](mailto:qhzhang@zju.edu.cn) (Q. Zhang).

<https://doi.org/10.1016/j.cej.2024.149929>

Received 22 November 2023; Received in revised form 23 January 2024; Accepted 21 February 2024

Available online 8 March 2024

1385-8947/© 2024 Published by Elsevier B.V.

occurrence of internal self-hydrolysis due to its strong hydrophilicity, significantly constraining the advancement of the system and results in considerable wastage of the antifouling agent [16,17].

Compared with other natural antifouling agents, eugenol has garnered significant attention for its cost-effectiveness and potent antifouling performance. And considerable researches have focused on exploring potential antifouling performance of eugenol in recent years [18]. For instance, Sha et al. successfully replaced metal ions in acrylate-based self-renewing coatings with eugenol, effectively mitigating the harmful effects associated with metal ion release [15]. Subsequently, they leveraged cation- $\pi$  interactions to develop a surface-anchored eugenol-acrylate system, further enhancing efficiency of eugenol [17]. However, both of them are prone to rapid hydrolysis and weight loss, challenging the construction of sustainable marine antifouling coatings. Combining eugenol with PDMS which is renowned for its exceptional weathering resistance and thermodynamic stability, offers a promising solution. As reported in 2021, a PDMS containing eugenol and coumarin demonstrated remarkable antifouling performance [19]. Nevertheless, the introduction of coumarin dimers with eugenol resulted in increased production costs and experimental complexity, impeding the industrialization of environmentally friendly antifouling coatings. Therefore, our study aims to develop a self-regulating, cost-effective marine antifouling coating utilizing eugenol as the sole antifouling agent. To address the aforementioned challenges, it is possible to seek inspiration from carp. When subjected to bacterial attacks, it secretes mucus that contains hydrophobic proteins which can make both gram-negative and gram-positive bacteria death [20]. This mucus serves as a protective barrier, “creating” a sterile environment in the vicinity of the fish. Taking inspiration from this self-protective mechanism of carp, a self-regulating coating was developed (Scheme 1) with robust controlled release of eugenol without internal self-hydrolysis. The self-renewal rate of the marine antifouling coating is feasibly controlled based on the hydrophilic of the coating as well as the concentration of the agent.

In this study, a novel fluorinated eugenol copolymer was synthesized and covalently grafted onto polyurea-modified PDMS to achieve the synthesis of a self-regulating coating (PDMS-Pun-x) with robust controlled release of eugenol. Under seawater conditions, the coating regulates its hydrophilicity and the content of the surface antifouling

agent, fulfilling controlled releasing eugenol and “establishing” an antifouling environment around the coating. Additionally, PDMS-Pun-x proves excellent self-healing performance and adhesive properties to the substrate. The coating exhibits a low surface energy (SE, 20–30 mJ/m<sup>2</sup>), low Young's modulus (<2.0 MPa), high self-healing rate (94.06 %), and a strong adhesion (>1.3 MPa). Furthermore, compared with commercial PDMS, the coating demonstrates remarkable antifouling performance, excellent corrosion resistance and robust weatherability. Therefore, this coating holds significant potential and promising application prospects for surface fouling prevention in marine equipment and devices.

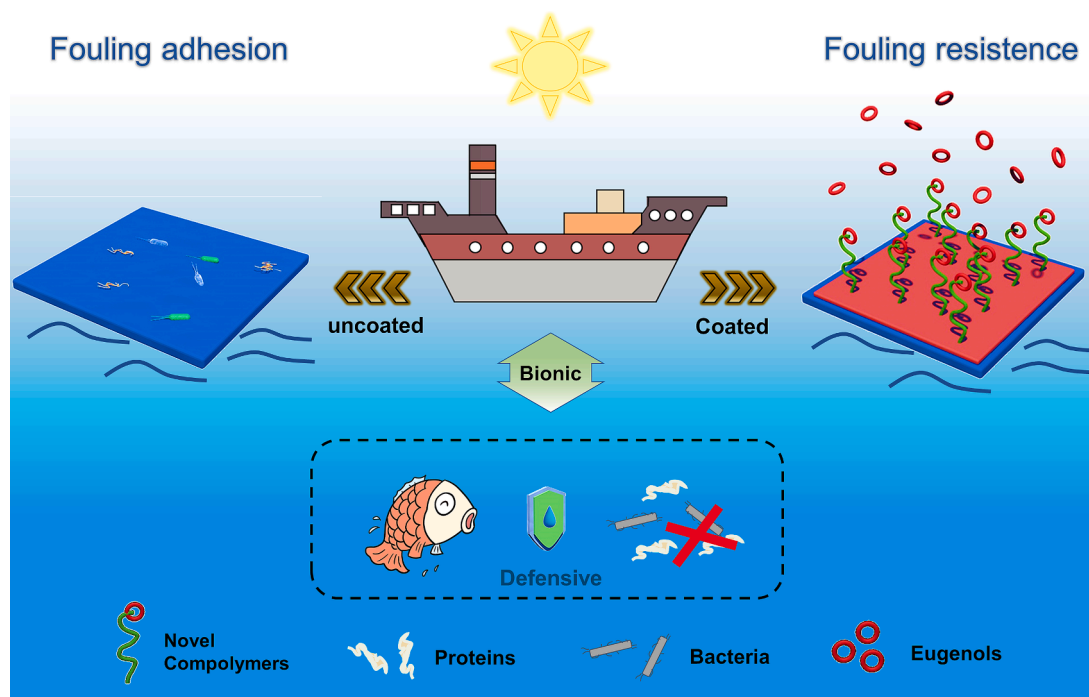
## 2. Results and discussion

### 2.1. Synthesis of eugenyl methacrylate (EnAc)

Eugenyl methacrylate (EnAc), as a novel eugenol-mimic monomer, was obtained by esterification of methacryloyl chloride and eugenol. A mixed solution of eugenol (1.7077 g, 10.4 mmol) and triethylamine (1.0524 g, 10.4 mmol) in dichloromethane (20 mL) was added dropwise methacryloyl chloride (0.9415 g, 10.4 mmol) under ice bath conditions and stirred for 0.5 h. The crude product solution was extracted with aqueous NaHCO<sub>3</sub> solution, water, and saturated brine and dried over anhydrous Na<sub>2</sub>SO<sub>4</sub>. The organic phase was evaporated and dried to give a yellow transparent liquid. The <sup>1</sup>H NMR spectrum of EnAc is illustrated in Fig. S3.

### 2.2. Synthesis of P(EnAc)-co-P(DFMA)

A typical protocol for the copolymerization of EnAc and 4-Aminodiphenyl sulfide (4,4'-DFMA) was conducted as follows. EnAc (1.09 g, 5.00 mmol), DFMA (2.00 g, 5.00 mmol), 3-Mercapto-1,2-propanediol (18.02 mg, 1.6 mmol), and AIBN (6.57 mg, 0.04 mmol) were weighed into a round-bottomed flask. Ultradry 20 mL tetrahydrofuran (THF) was added. The reaction solution was purged with nitrogen for 30 min. The sealed flask was immersed in an oil bath set at 70 °C for 14 h. After the completion of the reaction, the resulting solution in the flask was carefully transferred into 20 g of n-hexane to obtain a precipitate. Subsequently, the obtained precipitate was dissolved in THF and then



**Scheme 1.** Carp-inspired antifouling coating for ships with robust controlled release of eugenol.

subjected to another round of precipitation using n-hexane. This sequence of dissolution and precipitation was repeated three times, resulting in the formation of a milky-white, viscous liquid. The  $^1\text{H}$  NMR spectrum of P(EnAc)-co-P(DFMA) is illustrated in Fig. S4.

The narrow polydispersity index (PDI) of the copolymers confirms their repeatability, indicating excellent synthesis reproducibility (Table 1 and Fig. S5). Moreover, based on the principle that longer polymer chains promote the movement of segments to the surface [21], the copolymer with the highest molecular weight, P(EnAc)<sub>5</sub>-co-P(DFMA)<sub>5</sub>, was selected as the optimal fouling-release copolymer.

### 2.3. Synthesis of PDMS-Pun-x

Typically, 0.64 g (2.67 mmol) of Isophorone diisocyanate (IPDI) and 2.00 g (1.0 mmol) of hydroxypropyl terminated polydimethylsiloxane (HPT-PDMS) were added into a round-bottom flask with a condenser under the N<sub>2</sub> atmosphere with THF as solvent. The mixture was stirred at 75 °C for 3 h to yield a prepolymer. Subsequently, 0.201 g (0.067 mmol) of P(EnAc)<sub>5</sub>-co-P(DFMA)<sub>5</sub>, 0.101 g (0.407 mmol) of *ortho*-amino-phenylthioether (4,4'-DMPA), 0.009 g (0.856 mmol) of polyether amine (PEA) as the chain extender and 0.02 g of DBTDL were added and the reaction was conducted at 75 °C for 3 h. The polymer was dried under vacuum at 60 °C for 24 h. The dried polymer was dissolved in THF to use for coating. For convenience, the PDMS-based polyurea with eugenol-releasing performance (shown in Table 2) is designated as PDMS-Pun-x (x = 2.5, 5, 10), where x is the ratio of the total moles of P(EnAc)<sub>5</sub>-co-P(DFMA)<sub>5</sub> to the moles of HPT-PDMS. In addition, the commercial PDMS elastomer (Sylgard 184) as the control group was purchased from Dow Corning.

### 2.4. Structural characterization

Carp-inspired coating (PDMS-Pun-x) with controlled release of eugenol, excellent antibacterial properties, and long-term antifouling characteristics was obtained by reacting HPT-PDMS, IPDI, 4,4'-DMPA, PEA and P(EnAc)<sub>5</sub>-co-P(DFMA)<sub>5</sub>. The preparation route of PDMS-Pun-x is illustrated in Fig. 1a. FTIR tests were conducted on the PDMS-Pun-x. As depicted in Fig. 1b, the vibration peaks at 800 cm<sup>-1</sup> and 1008 cm<sup>-1</sup> correspond to the bending and stretching vibrations the Si-O-Si bond, respectively. The weakening and disappearance of the characteristic peak of NCO at 2270 cm<sup>-1</sup> and the appearance of CF at 1100 cm<sup>-1</sup> confirmed the successful synthesis of the coating.

The functionality and effectiveness of marine antifouling performance are significantly influenced by the properties of the surface. Therefore, EDS tests were conducted on PDMS-Pun-x (Fig. S6). The presence of six elements, namely C, N, O, S, Si, and F, on the surface of the coating confirmed the consistency between the actual situation and the designed structure of PDMS-Pun-x.

XPS analysis provided a detailed verification of the chemical composition on the surface. The surface elemental status of the initial PDMS-Pun-x was illustrated in Fig. 1c. The S 2p spectrum was deconvoluted into two component peaks, as shown in Fig. 1d, specifically S-S (168.00 eV) and S-C (163.2 eV). By deconvoluting the C 1 s, N 1 s, S 2p, O 1 s, and Si 2p spectra, the C 1 s spectrum was deconvoluted into five component peaks, as shown in the Fig. 1e. The CH<sub>x</sub> peak (including C-C, C-H) is located at 283.5 eV, while the other four peaks represent C-N/C-O/C-C-S (283.7 eV), C-Si (284.2 eV), C = O (285.2 eV), and CF<sub>2</sub> (288.00 eV). Similarly, the N 1s spectrum was deconvoluted into two component

**Table 1**  
Formulations of the novel copolymer,

| Polymers                                        | EnAc/DFMA/TPG  | M <sub>w</sub> (g/mol) | PDI   |
|-------------------------------------------------|----------------|------------------------|-------|
| P(EnAc) <sub>5</sub> -co-P(DFMA) <sub>5</sub>   | 1.00/1.00/0.33 | 3082                   | 2.742 |
| P(EnAc) <sub>2.5</sub> -co-P(DFMA) <sub>5</sub> | 1.00/2.00/0.5  | 2787                   | 2.701 |
| P(EnAc) <sub>1.7</sub> -co-P(DFMA) <sub>5</sub> | 1.00/3.00/0.67 | 1334                   | 1.528 |

**Table 2**  
Formulations of PDMS-Pun-x.

| Coatings     | PDMS(wt%)/Copolymer(wt%) | M <sub>w</sub> (g/mol) | PDI   |
|--------------|--------------------------|------------------------|-------|
| PDMS-Pun-2.5 | 100/2.5                  | 25,799                 | 3.008 |
| PDMS-Pun-5   | 100/5                    | 26,484                 | 3.610 |
| PDMS-Pun-10  | 100/10                   | 26,151                 | 3.622 |

peaks, as shown in Fig. 1f, namely N-H (398.79 eV) and C-N (398.10 eV). These findings further validate the successful preparation of PDMS-Pun-x and the migration of the copolymer to the coating surface, effectively preventing the failure of embedded antifouling agents.

The glass transition temperature (T<sub>g</sub>) of the PDMS-Pun-x coating was monitored by DSC testing (Fig. 1g). Typically, the T<sub>g</sub> is associated with the mobility of polymer segments, with lower T<sub>g</sub> indicating improved segment mobility and requiring less temperature and additional stimulation for self-healing. With an increase in the content of P(EnAc)<sub>5</sub>-co-P(DFMA)<sub>5</sub>, the T<sub>g</sub> of PDMS-Pun-x gradually increases while remaining below room temperature, indicating the coating's outstanding self-healing potential.

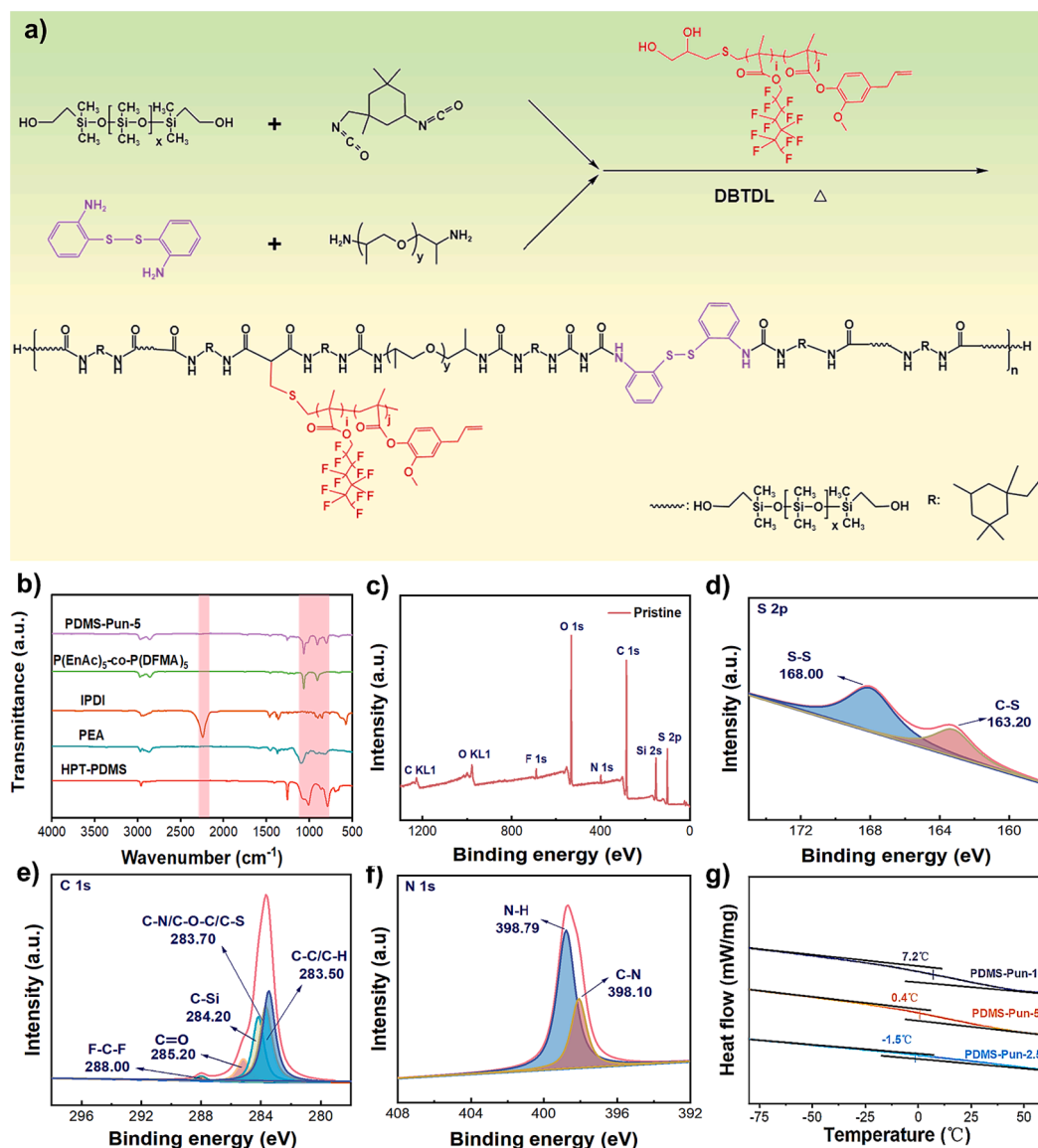
### 2.5. Hydrolyzing performances

Currently, the augmentation of the antifouling efficacy of conventional low surface energy (SE) coatings is accomplished by integrating physical doping agents. It is worth noting that although the method can enhance the antifouling performance to some extent, the physical doping coatings face challenges in achieving stable and controlled release of antifouling agents in complex marine environments.

To address this issue, researchers have explored alternative approaches to improve the antifouling efficacy of coatings. One such approach involves the incorporation of antifouling additives that can repel and hinder the approach of marine organisms toward the coatings. In the field of biomedicine, eugenol and its derivatives have gained recognition as effective antibacterial agents. These compounds are commonly utilized to prevent bacterial adhesion on various surfaces. Therefore, by incorporating eugenol into coatings, it is anticipated that significant advancements can be made in the field of marine antifouling coatings, offering a more environmentally friendly and sustainable solution to combat fouling.

P(EnAc)<sub>5</sub>-co-P(DFMA)<sub>5</sub>, an environmentally friendly eugenol-modified novel copolymer, allows for the slow release of eugenol through ester hydrolysis in weak alkaline seawater (pH≈8.0). To further evaluate the coating's ability to release eugenol in seawater, a 14-day static artificial seawater (ASW) experiment was conducted to verify the stable and persistent release of eugenol by the coating. Additionally, 7-day static alkaline hydrolysis experiment (pH = 11) was carried out to assess long-term antifouling potential of the coating.

The process of ester hydrolysis was investigated by monitoring the changes in coating wettability in ASW. The contact angles of the original coatings were all above 100° (Fig. 2a). Over time, the degree of hydrolysis of the coatings increased, causing the ester groups to transform into more hydrophilic carboxyl groups (Fig. S10), ultimately reaching a steady state. To simulate real-world scenarios, the hydrophobicity changes of the coatings immersed in ASW were evaluated by measuring the underwater oil contact angles (Fig. 2b). With an increase in copolymer content, the hydrophobicity of the coatings increased. As the hydrolysis time prolonged, a greater number of ester groups hydrolyzed into carboxyl groups, resulting in an increased hydrophilicity of the coatings. The SE of the coatings was determined by calculating the contact angle between CH<sub>2</sub>I<sub>2</sub> and WCA. As pointed out by the Bayes curve [22], when the coating possesses a lower surface energy (SE < 30 mJ/m<sup>2</sup>), it effectively reduces the interaction between organisms and the coating, inhibiting the attachment of barnacles. Moreover, the coatings exhibited self-cleaning properties, whereby attached organisms were easily removed by seawater when sufficient shear force is present.



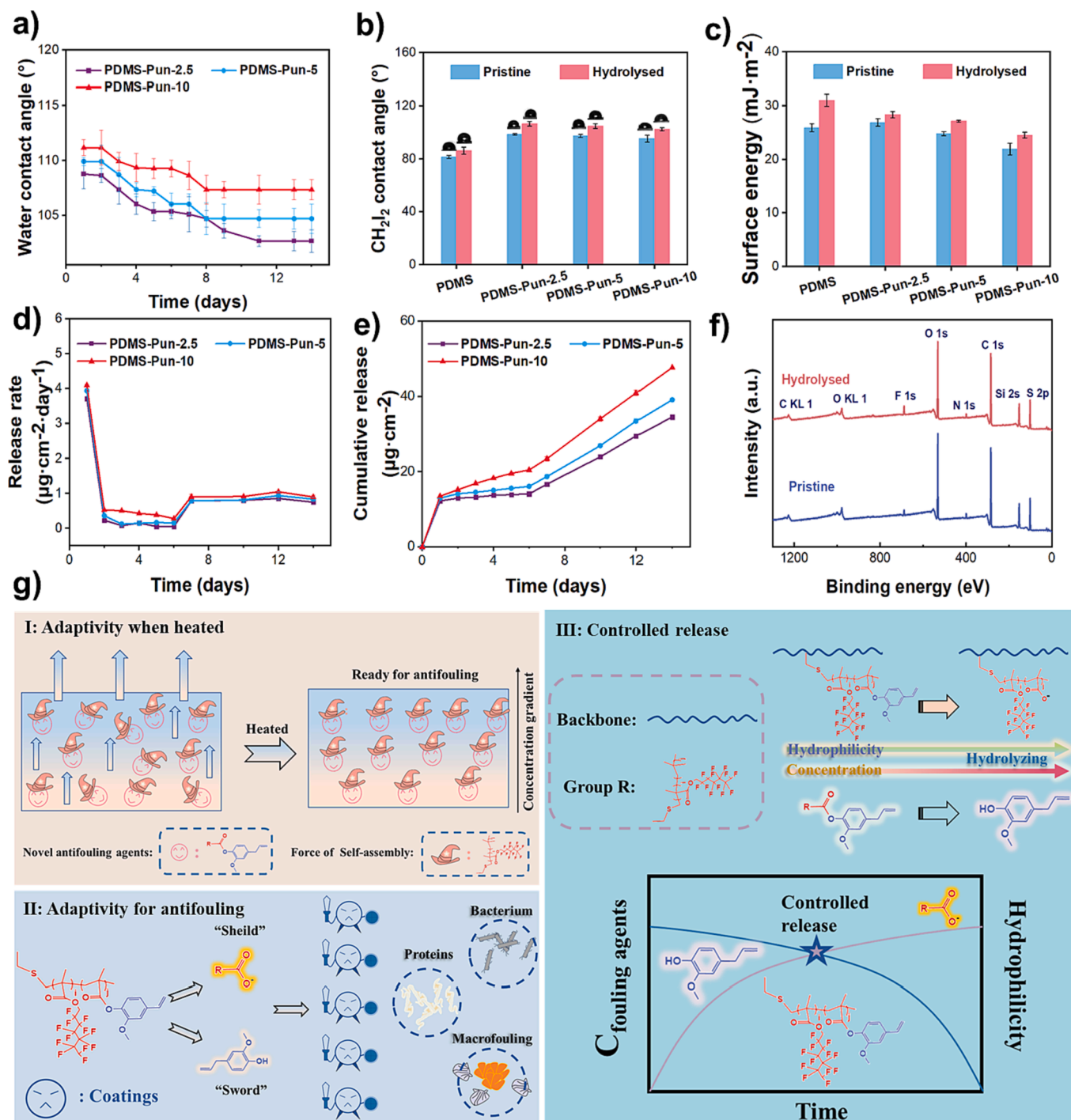
**Fig. 1.** (a) Synthesis process of antifouling coating. (b) FTIR spectra of PDMS-Pun-x and its ingredients. (c) XPS spectrum of PDMS-Pun-x. (d) S 2p spectrum of PDMS-Pun-x. (e) C 1s spectrum of PDMS-Pun-x. (f) N 1s spectrum of PDMS-Pun-x. (g) DSC of PDMS-Pun-x.

After immersion in seawater for 14 days, commercial PDMS experienced an increase in SE exceeding  $30 \text{ mJ}\cdot\text{m}^{-2}$ , thus causing a decrease in its self-cleaning performance. Conversely, the SE of PDMS-Pun-x remained below  $30 \text{ mJ}\cdot\text{m}^{-2}$  even after hydrolysis, indicating that they retain their pollution release capabilities (Fig. 2c).

The 14-day hydrolysis release rate and cumulative release amount of the coating in ASW were monitored (Fig. 2d and e). Eugenol was continuously and stably released by all coatings. The release rate of eugenol increased with a higher content of P(EnAc)<sub>5</sub>-co-P(DFMA)<sub>5</sub> in the coating. Due to the reconstructive effect of the seawater on the coating surface, a continuous decrease in the release rate of eugenol was observed. By the eighth day, the surface reconstruction of the coating was essentially completed (Fig. S12), resulting in the formation of numerous nanoscale micropores on the coating surface. This increased the contact area between the seawater and the coating, thereby accelerating the release rate of eugenol. Furthermore, XPS analysis was conducted on the hydrolyzed PDMS-Pun-x, revealing the presence of carbon (C), nitrogen (N), oxygen (O), sulfur (S), silicon (Si), and fluorine (F) elements (Fig. 2f). A noticeable increase in the fluorine content (Tables S1 and S2) was observed, indicating the remarkable surface self-

renewal capability of PDMS-Pun-x. This self-renewal capability is a desirable trait, as it can potentially enhance the durability and longevity of the coating. As a result, the methyl propyl eugenol ester on the coating surface was continuously replenished due to the self-migration effect of long-chain fluorine-containing. The coating achieved dynamic equilibrium with the seawater, enabling sustainable and stable release of eugenol eventually.

The hydrolysis rate of ester-based compounds is known to be dependent on the coating's hydrophilic or hydrophobic nature and the concentration of the antifouling agent [15]. In general, coatings with higher hydrophilicity and higher concentrations of the antifouling agent demonstrate an accelerated hydrolysis rate. Therefore, the robust controlled release properties of PDMS-Pun-x could be detailedly attributed to the exceptional adaptability of the coating (Fig. 2g), which can be delineated in three distinct steps. Initially, when the coating is forming, the antifouling agent continuously migrates towards the surface under the force of the fluorinated long chain, thus establishing a magic and dynamic concentration gradient distribution within the coating. Subsequently, in the presence of an alkaline marine environment, the coating undergoes hydrolysis to liberate eugenol, which assumes an



**Fig. 2.** (a) Water contact angle of PDMS-Pun-x at different immersion times. (b) CH<sub>2</sub>I<sub>2</sub> contact angle of PDMS-Pun-x and hydrolysed PDMS-Pun-x (underwater). (c) Surface energy change of PDMS-Pun-x and hydrolysed PDMS-Pun-x. (d) Release rate of eugenol from PDMS-Pun-x in seawater environment. (e) Cumulative release of eugenol from PDMS-Pun-x in seawater environment. (f) XPS spectra of PDMS-Pun-x before and after hydrolysis (compared with the pristine). (g) Mechanism of controlled release of eugenol.

active role in warding off fouling in the vicinity of the coating, much akin to a “sword.” Afterwards, the consequential formation of carboxylate anions augments the hydrophilicity of the coating, serving as a “shield” that restrains the adhesion of diverse hydrophobic fouling. As the result, the combination of sustained self-replenishment of the antifouling agent and the progressive enhancement of the coating’s hydrophilicity regulates the rate of eugenol hydrolysis and release together, effectively achieving controlled release of eugenol.

What’s more, the hydrolysis rate of esters is known to be influenced

by the nature of the hydrolysis group and the pH value [23,24]. To further evaluate the long-term ability of coatings in seawater, the hydrolysis release rate and cumulative release amount of eugenol-modified novel copolymer in a strong alkaline environment with a pH of 11 were monitored over a period of 7 days (Fig. S11). It was noted that eugenol was consistently and steadily released by all coatings. Similar to the hydrolysis behavior in ASW, the release rate of eugenol exhibited an upward trend with an increase in the content of P(EnAc)<sub>5</sub>-co-P(DFMA)<sub>5</sub> in the coating. However, due to the higher pH of the environment

compared with ASW under the same conditions, the coatings demonstrated a higher hydrolysis rate. Nonetheless, with prolonged hydrolysis time, the coating surface reached a dynamic equilibrium, enabling sustained and continuous release of eugenol for a duration of 7 days. This sustained release may be attributed to the forementioned explanation. Overall, these findings highlight the promising long-lasting antifouling potential of PDMS-Pun-x in alkaline seawater.

## 2.6. Antifouling performances

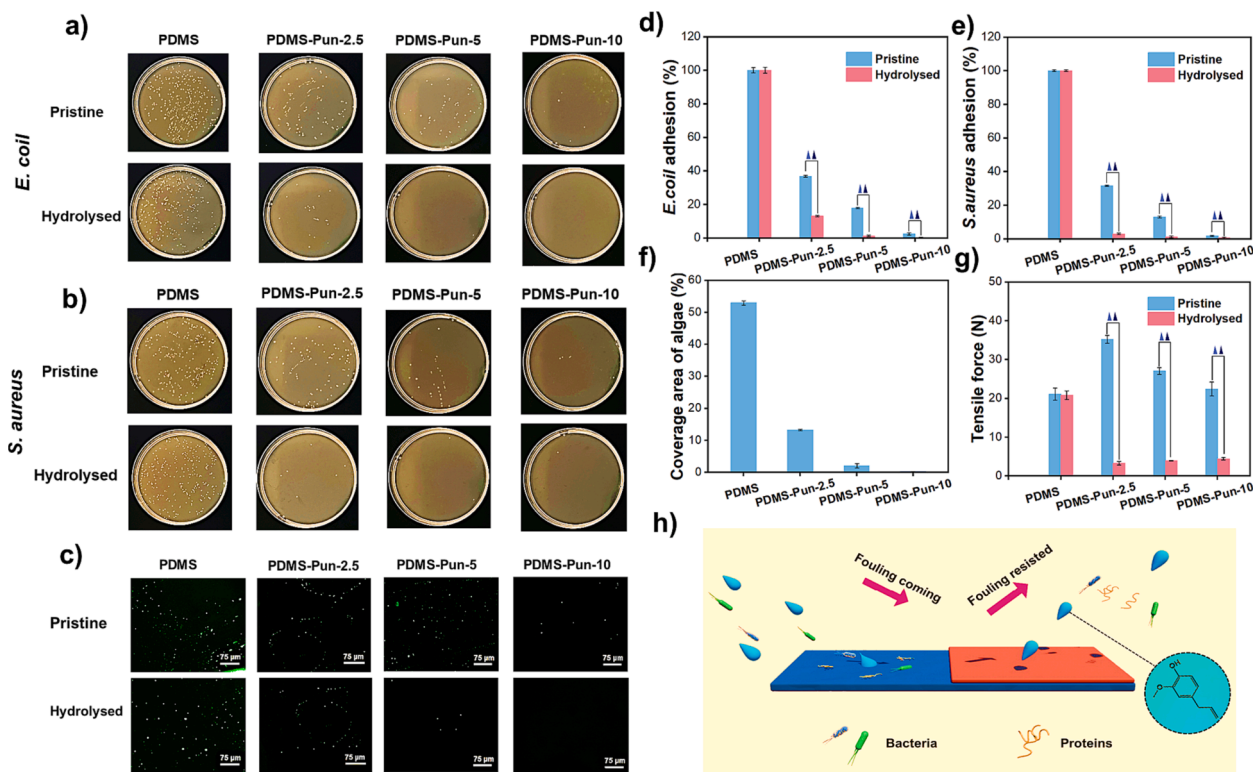
The ships navigating in the ocean are subjected to substantial contamination by bacteria [19,25–27]. To explore the antimicrobial properties of the coatings, tests were conducted using *Escherichia coli* (*E. coli*, a gram-negative bacterium), *Staphylococcus aureus* (*S. aureus*, a gram-positive bacterium), and bovine serum albumin (BSA). PDMS was utilized as the control group. The antifouling capabilities of PDMS-Pun-x before and after hydrolysis were evaluated (as shown in Fig. 3a–c). The observed results revealed a significant enhancement in the antibacterial and anti-protein efficiency of the hydrolyzed coating. Furthermore, PDMS-Pun-10 exhibited an impressive inhibition efficiency of over 95 % against both *E. coli* and *S. aureus*, underscoring a notable improvement in antibacterial effects compared to commercially available PDMS coatings (Fig. 3d and e).

The bacteriostatic mechanism can be summarized as follows (shown in Fig. 3h): Firstly, the phenoxy anion dissociates from the phenol group and forms a chemical bond with the active site of the membrane protein on the bacterial cell membrane, thereby disrupting the structure and function of the membrane [28]. The activity of malate dehydrogenase and succinate dehydrogenase are inhibited by eugenol, impeding the REDOX process of the tricarboxylic acid cycle (TCA cycle) during bacterial respiration [19]. Secondly, the addition of fluorine-containing copolymers further reduces the SE of the coating, enhancing its

resistance to adhesion. Additionally, the coating's microphase separation characteristics and the heterogeneity of its molecular scale composition enable the establishment of unfavorable interactions with pollutants [29], effectively preventing or minimizing adsorption events. Lastly, the hydrolyzed coating possesses an abundance of carboxyl groups, enhancing the hydrophilicity of the coating and further reinforcing its resistance to adhesion. The strong affinity between the cell wall of diatoms and the main constituents of PDMS, such as silicon dioxide, is the primary factor contributing to the extensive adhesion of diatoms onto PDMS coatings. As for fouling formed by diatoms, PDMS-Pun-x, featuring a microphase-separated structure, exhibits reduced diatom adhesion (Figs. 3f and S13). The hydrolysis-induced formation of carboxyl groups enhances the hydrophilicity of the coating, thereby augmenting its defense performance against diatom attachment. Concurrently, the inclusion of fluorine-containing segments in the copolymer promotes the release of fouling agents, facilitating the removal of adhered substances. Besides, the eugenol released during hydrolysis effectively impedes the growth of diatoms on PDMS-Pun-x surface [30], employing mechanisms akin to antibacterial and anti-protein activities.

The fouling of barnacles, a common large marine organism, was evaluated using aluminum barnacles to assess the fouling release capability of PDMS-Pun-x [19]. The hydrolyzed PDMS-Pun-x demonstrated significant reduction in adhesion, particularly PDMS-Pun-10, which required only 4.43 N for removal, representing a 79 % decrease compared with PDMS. This confirms that flowing seawater can easily remove barnacles during the ship's navigation process (Fig. 3g).

Marine fouling encompasses not only organic fouling caused by marine organisms on the coating surface, but also the attachment of sediments. Therefore, hydrophilic copper chloride dihydrate was employed as the hydrophilic inorganic fouling model, while silica particles served as the hydrophobic inorganic fouling model, to conduct



**Fig. 3.** (a) *E. coli* with different coatings incubated for 12 h. (b) *S. aureus* with different coatings incubated for 12 h. (c) Representative fluorescence microscope images before and after hydrolysis of PDMS-Pun-x. (d) *E. coli* adhesion rate of PDMS-Pun-x before and after hydrolysis under plate counting method. (e) *S. aureus* adhesion rate of PDMS-Pun-x before and after hydrolysis under plate counting method. (f) Percentage of the covered area of diatoms before and after hydrolysis. (g) Removal power of barnacles attached to different coatings before and after hydrolysis. (h) Antibacterial mechanism of PDMS-Pun-x.

self-cleaning experiments on PDMS-Pun-x. The fouling particles were positioned at a 45° on the sample surface, and deionized water was introduced above the coating to remove the inorganic fouling model. The results, depicted in Fig. S14, illustrated that the majority of the inorganic fouling did not adhere to PDMS-Pun-x, as it was effectively flushed away by the water flow, which highlights the self-cleaning performance of PDMS-Pun-x.

## 2.7. Mechanical property and self-healing ability

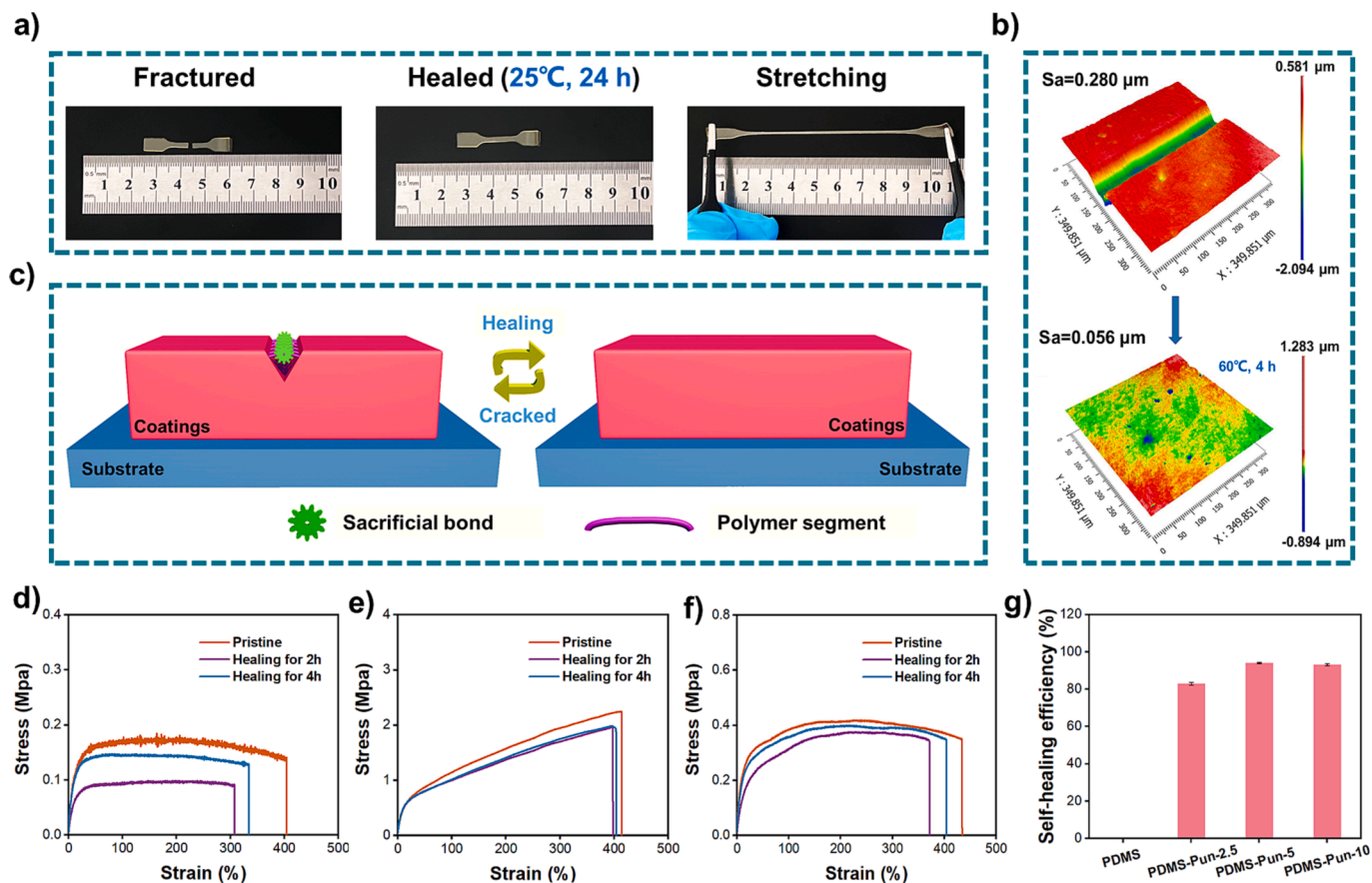
Inherent defects in the mechanical properties of traditional PDMS coatings result in unavoidable irreversible deformation and damage during usage, thereby restricting its applications in various fields [29,31]. PDMS-Pun-x are composed of molecular chains comprising soft segments containing siloxane and ether bonds, as well as hard segments containing amino-formate and urea bonds. The flexible soft segments with low glass transition temperature form the fundamental phase, while the hard segments aggregate to form the hard segment portion of the polyurea under the physical crosslinking effect of a high cohesive energy derived from polar groups [32]. Consequently, PDMS-Pun-x exhibits excellent physical and chemical properties. The mechanical properties of PDMS-Pun-x are closely correlated with the content and structure of the hard and soft segments. The relationship between stress and the content of hard segments in PDMS-Pun-x, with the same structure, is demonstrated in Fig. S15a. Among the PDMS-Pun-x, PDMS-Pun-5 demonstrates the optimal mechanical properties, featuring a lower elastic modulus of 0.86 MPa (shown in Fig. S15b). During the stress-strain tests, PDMS-Pun-5 exhibits a stress tolerance of 2.2 MPa (1.63 times higher than that of commercial PDMS) and a strain capacity 6.09

times greater than that of PDMS, reaching 429.2 %. These enhancements are attributed to the incorporation of side chains with multiple phenyl rings in the polymer chains, which enhance the rigidity of the coatings. Additionally, the physical crosslinking effect between polymer chain segments is facilitated by hydrogen bonds, thereby augmenting the stress resistance of the coating. The inclusion of polyether amine segments with ether linkages effectively mitigates stress for polymer segments [33,34], thereby enhancing the flexibility of the polymer chains and consequently improving the strain capability of the polymer chain.

Based on the findings from the previous DSC test, it can be deduced that PDMS-Pun-x exhibits a lower T<sub>g</sub>, with all measured values below the standard room temperature of 25 °C. The observation indicates the migratory nature of the molecular chains of the soft segments in the coating at room temperature.

The self-healing process of the dumbbell-shaped coating at room temperature was recorded (Fig. 4a), revealing that the coating demonstrates effective self-healing behavior within 24 h at room temperature. In addition, the coating exhibited a remarkable self-healing ability even in the presence of ASW (Fig. S18), which underscored their potential to effectively repair themselves in marine environments below room temperature. And such a characteristic not only highlighted the ability to withstand complex marine conditions but also presents a promising avenue for enhancing the mechanical performance of PDMS. In fact, the increase in temperature facilitates the enhancement of the chain's migratory ability. Subsequently, a temperature of 60°C was applied to the coating to elucidate the interrelationship between the self-healing rate and self-healing time.

The self-healing mechanism of PDMS-Pun-x is illustrated in Fig. 4c, demonstrating three potential healing modes: (i) at room temperature,

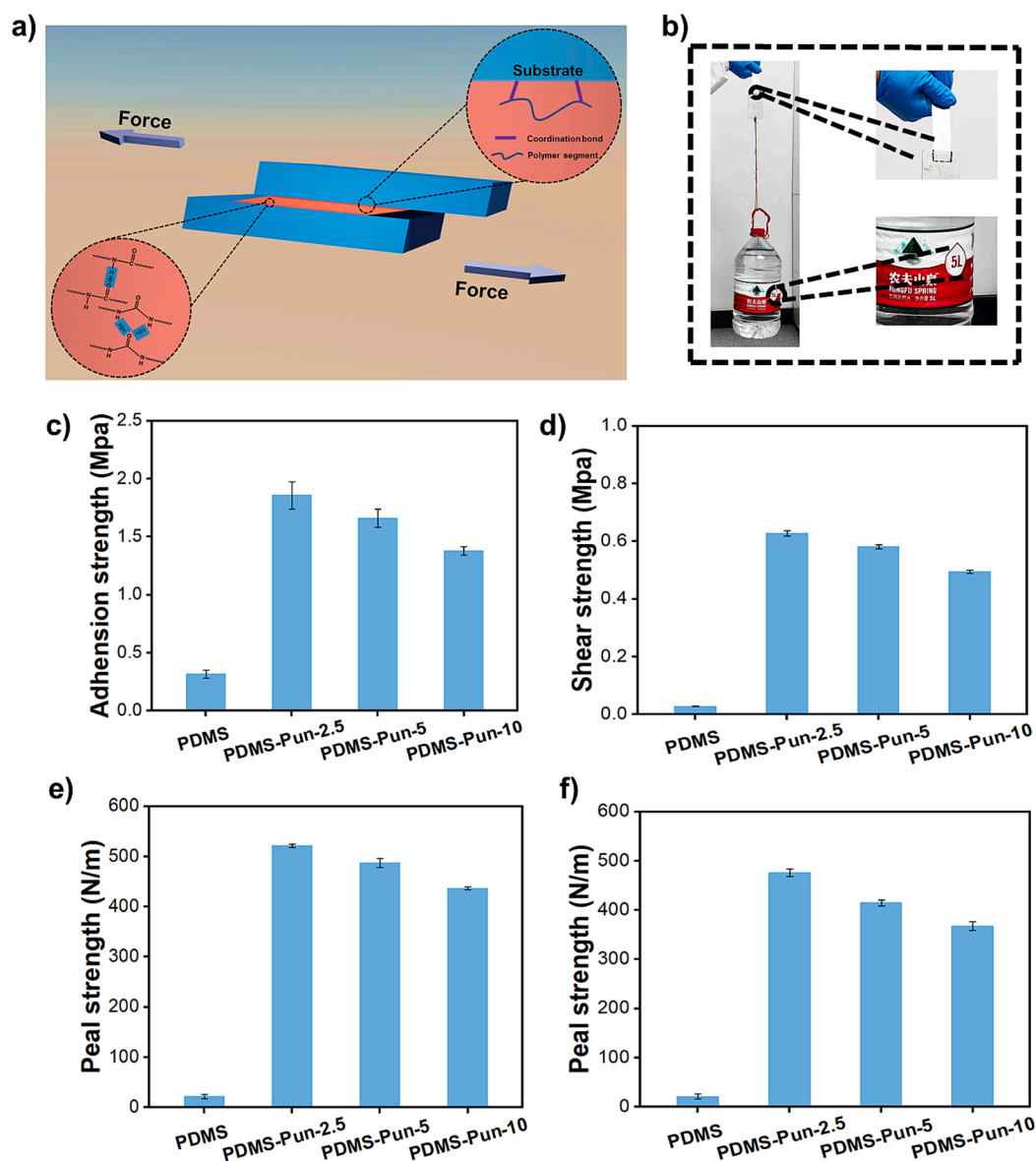


**Fig. 4.** (a) Picture showing the cut-off and healed PDMS-Pun-5 sample. (b) Roughness change of PDMS-Pun-x before and after self-healing (scratch type). (c) Schematic diagram of self-healing mechanism. (d) Stress-strain curve of PDMS-Pun-2.5 (pristine and healing for 2 h, 4 h). (e) Stress-strain curve of PDMS-Pun-5 (pristine and healing for 2 h, 4 h). (f) Stress-strain curve of PDMS-Pun-10 (pristine and healing for 2 h, 4 h). (g) Self-healing efficiency of PDMS and PDMS-Pun-x.

physical crosslinking is induced through the interaction of dihydrogen bonds; (ii) at 60 °C, the disulfide bonds undergo a reverse reaction, enabling small molecules and low molecular weight molecules to fill the gaps, resulting in tightly intertwined molecular chains; (iii) at 60 °C, the healing of the interface is initiated through the synergistic effect of disulfide bonds and hydrogen bonds (both sacrificial bonds). To investigate the self-healing characteristics and healing mechanism of PDMS-Pun-x, the self-healing efficiency was evaluated through tensile experiments. The samples were cut into different segments and subjected to 60 °C for healing. Fig. 4d-f present the stress-strain curves of PDMS-Pun-2.5, PDMS-Pun-5, and PDMS-Pun-10 after 2 h and 4 h. It is observed that a longer healing time corresponds to a higher self-healing rate. This phenomenon could be attributed to the increased mobility and disruption of hydrogen bonds at elevated temperatures [34–36]. The mechanically cleaved mercapto radicals generated by activated disulfide bonds rapidly exchange with neighboring disulfide bonds, leading to the formation of new disulfide bonds. Additionally, higher temperatures and longer healing times facilitated the faster diffusion of molecular chains into the damaged region, enabling efficient self-healing of PDMS-Pun-x. Among these samples, PDMS-Pun-5 exhibited a remarkable healing rate

of 94.06 % after 4 h. Moreover, PDMS-Pun-5 demonstrated the highest recovery of fracture elongation at the same healing time, indicating an optimal balance between fracture elongation and self-healing rate.

Furthermore, standard techniques, as described in reference [35], were employed to evaluate the healing behavior of the coating (shown in Fig. S19). Scratch healing tests were conducted to assess the physical and chemical properties of the coating affected by the roughness of the polyurea. Therefore, during the evaluation of the healing behavior of the coating, roughness tests were performed using a laser profilometer (shown in Figs. 4b and Fig. S16). The results indicated that the roughness of the coating did not exhibit significant changes compared to the initial coating, suggesting that the coating maintained good stability during the self-healing process without any noticeable decomposition. Moreover, it is worth noting that surface roughness plays a significant role in the attachment of marine organisms. Generally, a rough surface provides more attachment sites, promoting the growth of fouling organisms and exacerbating fouling. The scratch healing test of the coating revealed that the healed coating still maintained a low surface roughness ( $<0.1 \mu\text{m}$ ) [37], confirming the coating's great ability to effectively detach fouling materials. Additionally, recyclable experiments were



**Fig. 5.** (a) Adhesion enhancement mechanism. (b) Adhesion diagram of PDMS-Pun-5. (c) Adhesion force of the coating. (d) Shear strength of the coating. (e) 180° peeling force of the coating. (f) 90° peeling force of the coating.

conducted to verify the outstanding recyclable properties of the carp-inspired coating (Fig. S22). These results unequivocally demonstrated the practicality and utility of the coating, emphasizing its potential for practical applications.

## 2.8. Adhesion test

Given the crucial role of adhesion strength between the coating and the substrate in determining the longevity of the coating, a series of comprehensive adhesion tests were undertaken to assess this critical parameter. The mechanism of the improvement in adhesion strength of the coating is illustrated in Fig. 5a.

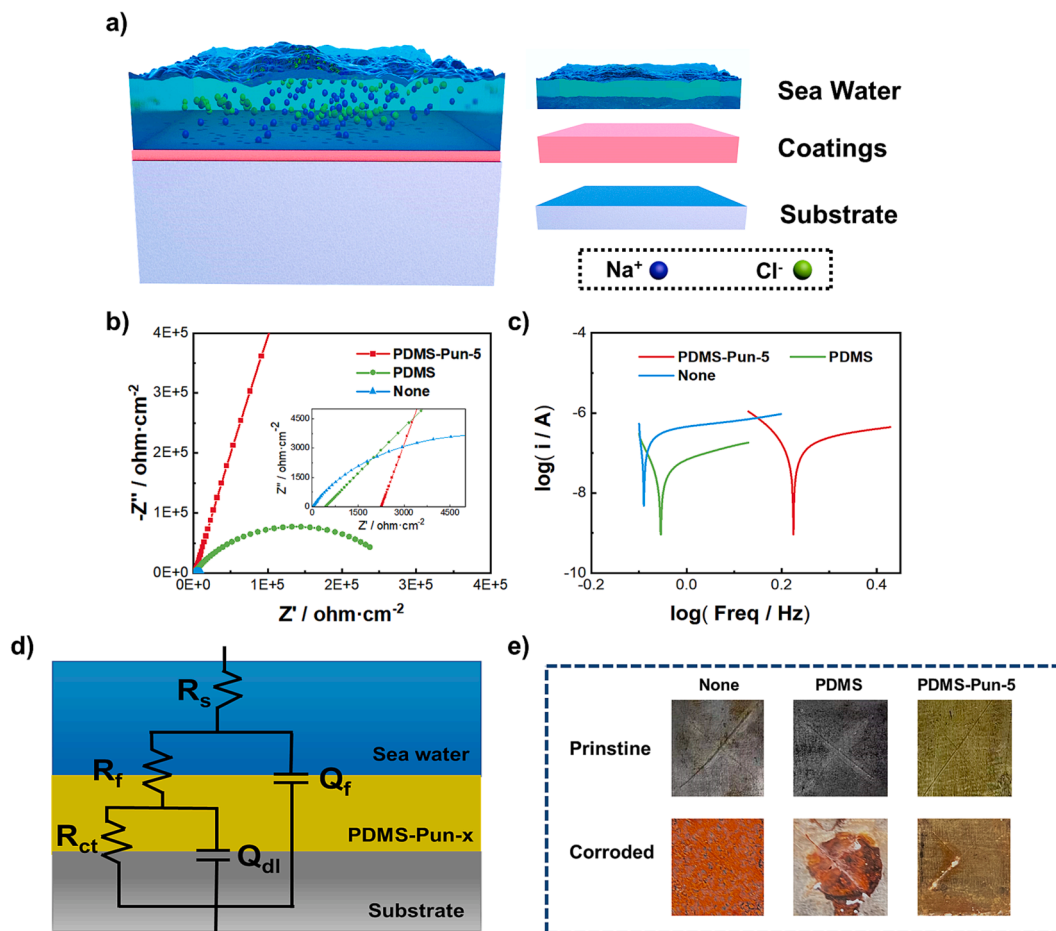
The increase in adhesion strength can primarily be ascribed to two key factors: the introduction of covalent bonding between EnAc and DFMA, which imparts the cured coating with an abundance of ester groups capable of forming coordinating interactions with the metal substrate; and the structural presence of carbamate and urea bonds within the coating, which endow the coating with a rich array of hydrogen bonds. On one hand, the formation of hydrogen bonds within the system acts as physical cross-links during the coating formation process, thereby strengthening the mechanical integrity of the coating. On the other hand, the increased hydrogen bonds enhance the inter-molecular interactions between polymer segments [38], resulting in higher resistance to segment separation and ultimately enhancing the adhesive performance of the coating. Fig. 5c confirms the significant improvement in the adhesion performance of the coating. It is observed that the adhesion strength between PDMS-Pun-x and the steel plate surpasses that of PDMS (0.315 MPa), with a maximum value of 1.85 MPa.

Furthermore, as the copolymer content increases, the adhesion strength of the coating diminishes. This reduction can be attributed to the decrease in carbamate and urea bonds due to the higher copolymer content, resulting in a lower hydrogen bond density and subsequently reducing the adhesion strength.

The adhesion performance of the coating to the substrate could also be experimentally verified through lap shear testing (Fig. 5d). It was observed that the lap shear strength of PDMS-Pun-x can reach a maximum of 0.627 MPa, which was 22.8 times higher than that of the PDMS coating (0.027 MPa). The adhesion performance of the coating was further evaluated using 180° and 90° peel tests (Fig. 5e, f). The gradual reduction in the peel force was observed with an increase in copolymer content, although the peel force remained significantly higher than that of commercially available PDMS coatings. At the optimum ratio, the adhesion strength of PDMS-Pun-2.5 reached 475 N/m, which is substantially greater than the 21.27 N/m of PDMS. Lastly, a macroscopic demonstration was conducted to showcase the adhesion strength of the coating, where a 0.25 g coating proved capable of lifting 5L of water (Fig. 5b). These findings provide substantial evidence that the covalent introduction of carbamate and urea bonds has greatly enhanced the adhesion performance of the coating.

## 2.9. Anti-corrosion tests

The detrimental role of seawater corrosion poses substantial limitations on the application and utilization of coatings in marine environments [39]. Polyurea coatings exhibit exceptional corrosion resistance properties [40]. To validate the anti-corrosion performance of PDMS-



**Fig. 6.** (a) Anti-corrosion diagram of PDMS-Pun-x. (b) Measurement of  $\log|Z|$  under different coating conditions (ASW). (c) Measurement of Tafel curve under different coating conditions (ASW). (d) Equivalent circuit diagram with coating protection (ASW). (e) Corrosion image of different cases after 14 days (ASW).

Pun-x, electrochemical techniques such as electrochemical impedance spectroscopy (EIS) were employed. The impedance is represented by the equivalent circuit diagram presented in Fig. 6d. The circuit elements encompass solution resistance ( $R_s$ ), polyurea coating resistance ( $R_f$ ), and charge transfer resistance ( $R_{ct}$ ), as well as the double-layer capacitance of the polyurea ( $Q_f$ ) and the steel substrate ( $Q_{dl}$ ).

As indicated by the Nyquist plot in Fig. 6b, it could be concluded that at low frequencies, the semi-circle of PDMS-Pun-5 was larger than that of PDMS, and the latter was larger than that of the uncoated substrate. The charge transfer resistance ( $R_{ct}$ ) of the polyurea coating was measured to be  $9.41 \times 10^6 \Omega \cdot \text{cm}^2$ , which was 187 times higher than that of PDMS ( $5.0349 \times 10^4 \Omega \cdot \text{cm}^2$ ), significantly surpassing the  $R_{ct}$  of the uncoated substrate. This suggested that when the steel substrate lacked protective coating, the corrosion of the coating become severe (Fig. 6e), while the polyurea coating ensured a considerably sluggish charge transfer process [41]. Additionally, tafel plots serve as vital criteria for evaluating the corrosion resistance performance of coatings. The Tafel plot of the coated steel electrode was recorded after immersing it in seawater for 14 days. From Fig. 6c, it could be inferred that PDMS-Pun-x exhibited the lowest corrosion current, followed by PDMS, while the uncoated substrate displays the highest corrosion current. Regarding the corrosion potential ( $E_{\text{corr}}$ ) value, PDMS-Pun-x was positioned at the far right, further validating its exceptional corrosion resistance [42]. The outstanding corrosion resistance mechanism of PDMS-Pun-x could be attributed to two aspects (Fig. 6a) [43]. Firstly, the hydrophobic characteristics of the coating, along with the microphase separation leading to wrinkled surface morphology, were confirmed by SEM images and EIS results. This structure facilitated the entrapment of air, leading to a reduced actual contact area with water. Consequently, corrosive ions were effectively hindered from participating in electrochemical reactions [44]. Secondly, the physical crosslinking formed by hydrogen bonding between polymer chains contributed to the formation of a dense polyurea layer. In seawater, water molecules often existed in an associated state, forming large clusters through hydrogen bonding. These clusters, with diameters larger than 20 nm, found it difficult to penetrate the gaps between polyurea molecules. As a result, when the polyurea molecules filled the micropores and gaps, the coating exhibited excellent corrosion resistance. Fig. 6e shows the corrosion conditions observed under different coating scenarios, including PDMS-Pun-x, PDMS coating, and unprotected surfaces, demonstrating that the polyurea

matrix in seawater undergoes minimal chemical changes, exactly indicating its potential as a long-lasting marine antifouling coating.

## 2.10. Chemical stability

Herein, separate evaluations were conducted on PDMS-Pun-x for acid resistance ( $\text{pH} = 3$ ). Additionally, owing to the remarkable self-healing capability and mechanical properties of PDMS-Pun-5, the specific ratio was chosen to subject it to further tests, such as high temperature endurance ( $120^\circ\text{C}$ ), low temperature endurance ( $-5^\circ\text{C}$ ), and UV resistance, in order to completely assess its robust stability in extreme environmental conditions. Fig. 7a shows that, after being immersed in acid ( $\text{pH} = 3$ ) for 7 days, the contact angle of PDMS-Pun-x slightly decreased, and the SE slightly increased as well. This could be attributed to the prevention of acid penetration by the PDMS-Pun-x segments and the excellent acid resistance of the polyurea coating [45]. It was also observed that the microstructure of the coating did not undergo any significant changes (Fig. S25b).

The results of the high temperature resistance ( $120^\circ\text{C}$ ), low temperature resistance ( $-5^\circ\text{C}$ ), and UV resistance tests, as depicted in Fig. 7b, showed that the contact angle of the coating fluctuated within different ranges as the exposure time to extreme environments increased. However, it still remained around the initial value of  $109.923^\circ \pm 0.371^\circ$ . Surface roughness tests (Fig. 7c) indicated that the coating maintained good roughness ( $0.1\text{--}0.16 \mu\text{m}$ ) even in extreme environments, which helped reduce drag and the attachment of dirt and organisms. Therefore, the coating exhibited excellent chemical stability and has the potential to be used in marine equipment operating in various extreme environments.

## 3. Conclusion

In conclusion, inspired by the self-protective mechanism exhibited by carp, this study introduces the development of a novel self-regulating marine antifouling coating (PDMS-Pun-x). Under seawater, the coating regulates of its inherent hydrophilicity and the content of the surface antifouling agent to obtain the ability of controlled releasing eugenol. In comparison to other self-renewal coatings, it demonstrates an absence of heavy metal leaching and internal self-hydrolysis, setting it apart as a superior alternative in terms of environmental safety and longevity. This

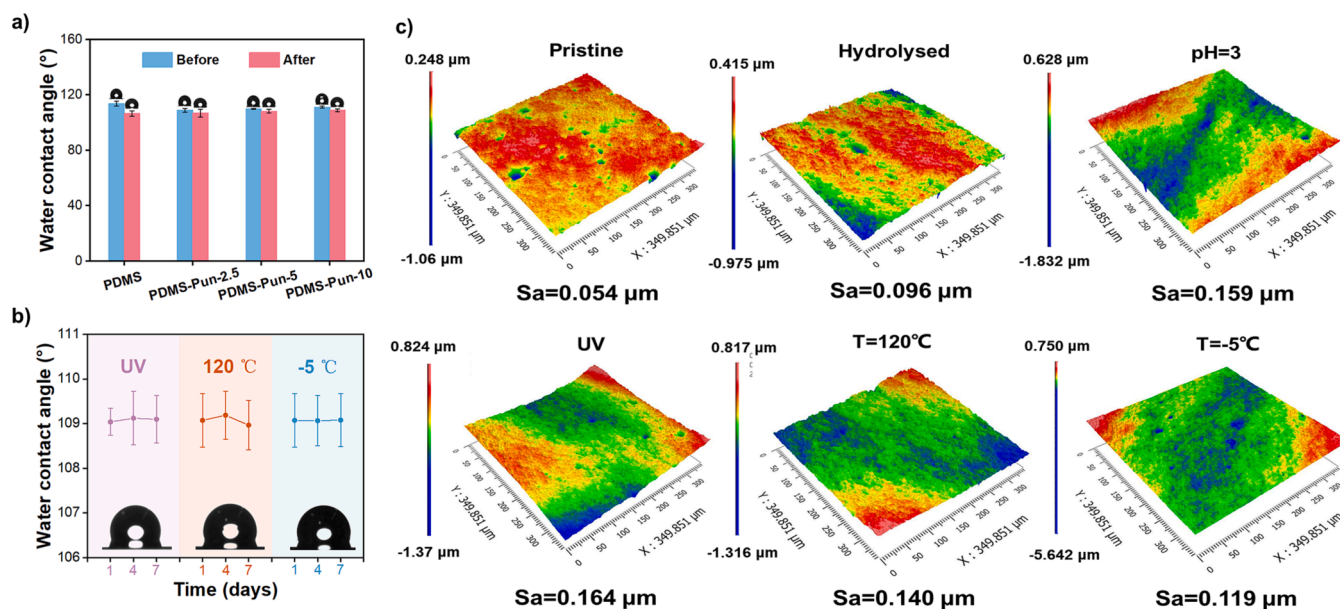


Fig. 7. (a) PDMS-Pun-5 SEM image after 7 days immersed in acid solution ( $\text{pH} = 3$ ). (b) Water contact angle of PDMS-Pun-x under different environments (UV,  $120^\circ\text{C}$ ,  $-5^\circ\text{C}$ ). (c) Roughness of PDMS-Pun-x in different cases.

research not only addresses the issue of limited static antifouling ability in PDMS coatings but also provides an innovative approach to reduce self-renewal coatings' reliance on hydrodynamic shear forces. As a result, this study offers potential applications in various fields, including biomedicine and marine antifouling.

### CRediT authorship contribution statement

**Jiahuan Liu:** Writing – review & editing, Writing – original draft, Data curation, Conceptualization. **Qiang Li:** Visualization. **Taijiang Gui:** Project administration, Data curation. **Hongyu Guo:** Writing – review & editing. **Kai Zhang:** Validation. **Fandong Meng:** Formal analysis. **Xiaoli Zhan:** Supervision, Conceptualization. **Quan Liu:** Supervision, Resources. **Qinghua Zhang:** Conceptualization, Funding acquisition, Methodology.

### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

### Data availability

Data will be made available on request.

### Acknowledgements

This work was supported by the National Natural Science Foundation of China (22308320), Research Funds of Institute of Zhejiang University-Quzhou (IZQ2021RCZX037, ACTIC-2022-008 and IZQ2022KJ2002).

### Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cej.2024.149929>.

### References

- H. Qiu, K. Feng, A. Gapeeva, K. Meuris, S. Kaps, X. Li, L. Yu, Functional polymer materials for modern marine biofouling control, *Prog. Polym. Sci.* 127 (2022) 101516, <https://doi.org/10.1016/j.progpolymsci.2022.101516>.
- H. Jin, L. Tian, W. Bing, J. Zhao, L. Ren, Bioinspired marine antifouling coatings: status, prospects, and future, *Prog. Mater. Sci.* 124 (2022) 100889, <https://doi.org/10.1016/j.pmatsci.2021.100889>.
- Z. Tong, L. Song, S. Chen, J. Hu, Y. Hou, Hagfish-inspired Smart SLIPS marine antifouling coating based on supramolecular: lubrication modes responsively switching and self-healing properties, *Adv. Funct. Mater.* 32 (27) (2022) 2201290, <https://doi.org/10.1002/adfm.202201290>.
- H. Guo, L. Song, J. Hu, T. Lin, X. Li, H. Yu, D. Cheng, Y. Hou, X. Zhan, Q. Zhang, Enhanced antifouling strategy with a strong synergistic effect of fluorescent antifouling and contact bacteriostasis using 7-amino-4-methylcoumarin, *Chem. Eng. J.* 420 (2021) 127676, <https://doi.org/10.1016/j.cej.2020.127676>.
- Z. Tong, H. Guo, Z. Di, S. Chen, L. Song, J. Hu, P. pavoninus-Inspired Smart Slips Marine Antifouling Coating Based on Coumarin: Antifouling Durability and Adaptive Adjustability of Lubrication, *Adv. Funct. Mater.* 23 (2023) 10702, <https://doi.org/10.1002/adfm.202310702>.
- Z. Tong, F. Gao, S. Chen, L. Song, J. Hu, Y. Hou, J. Lu, Slippery porous-liquid-infused porous surface (SPIPS) with on-demand responsive switching between “defensive” and “offensive” antifouling modes, *Adv. Mater.* 23 (2023) 08972, <https://doi.org/10.1002/adma.202308972>.
- J. Pan, Q. Xie, H. Chiang, Q. Peng, P.-Y. Qian, C. Ma, G. Zhang, “From the nature for the nature”: an eco-friendly antifouling coating consisting of poly (lactic acid)-based polyurethane and natural antifoulant, *ACS Sustain. Chem. Eng.* 8 (3) (2020) 1671–1678, <https://doi.org/10.1021/acssuschemeng.9b06917>.
- H. Zhang, Y. Li, S. Tian, X. Qi, J. Yang, Q. Li, C. Lin, J. Zhang, L. Zhang, A switchable zwitterionic ester and capsaicin copolymer for multifunctional marine antibiofouling coating, *Chem. Eng. J.* 436 (2022) 135072, <https://doi.org/10.1016/j.cej.2022.135072>.
- D.B. Ramsay, G.H. Dickinson, B. Orihuela, D. Rittschof, K.J. Wahl, Base plate mechanics of the barnacle *Balanus amphitrite* (=Amphibalanus amphitrite), *Biofouling*. 24 (2) (2008) 109–118, <https://doi.org/10.1080/08927010701882112>.
- L. Tian, J. Wang, Y. Yin, W. Bing, W. Du, H. Jin, Coffee/polydimethylsiloxane composite coating for preventing marine biofouling, *J. Coat. Technol. Res.* 20 (3) (2022) 949–955, <https://doi.org/10.1007/s11998-022-00714-x>.
- S. Zhao, D. Zhang, A silicone coating containing natural borneol fluorinated side chains with excellent static antifouling properties, *Eur. Polym. J.* 193 (2023) 112064, <https://doi.org/10.1016/j.eurpolymj.2023.112064>.
- C. Ma, W. Xu, Degradable polymers for marine antibiofouling: optimizing structure to improve performance, *Ind. Eng. Chem. Res.* 55 (44) (2016) 11495–11501, <https://doi.org/10.1021/acs.iecr.6b02917>.
- Y. Han, W. Zhao, Y. Zheng, H. Wang, Y. Sun, Y. Zhang, J. Luo, H. Zhang, Self-adhesive lubricated coating for enhanced bacterial resistance, *Bioact. Mater.* 6 (8) (2021) 2535–2545, <https://doi.org/10.1016/j.bioactmat.2021.01.028>.
- Q. Xie, Q. Xie, J. Pan, C. Ma, G. Zhang, Biodegradable polymer with hydrolysis-induced zwitterions for antibiofouling, *ACS Appl. Mater. Interfaces*. 10 (13) (2018) 11213–11220, <https://doi.org/10.1021/acsami.8b00962>.
- J. Sha, J. Yu, R. Chen, Q. Liu, J. Liu, J. Zhu, Eco-friendly self-polishing antifouling coating via eugenol ester hydrolysis, *Prog. Org. Coat.* 172 (2022) 107077, <https://doi.org/10.1016/j.porgcoat.2022.107077>.
- W. Zhou, Y. Zhou, C. Ni, L. Yu, C. Li, X. Yan, X. Zhang, Research on electromagnetic wave absorption properties of zinc-based acrylate resins for marine antifouling coating, *J. Alloys Compd.* 900 (2022) 163285, <https://doi.org/10.1016/j.jallcom.2021.163285>.
- J. Sha, X. Liu, R. Chen, J. Yu, Q. Liu, J. Liu, J. Zhu, Surface hydrolysis-anchored eugenol self-polishing marine antifouling coating, *J. Colloid Interface Sci.* 637 (2023) 67–75, <https://doi.org/10.1016/j.jcis.2023.01.058>.
- M. García, M. Stupak, M. Pérez, G. Blustein, Transitioning to nontoxic antifouling paints, *Pigm. Resin Technol.* 44 (2) (2015) 116–121, <https://doi.org/10.1108/prt-03-2014-0022>.
- Z. Tong, Q. Rao, S. Chen, L. Song, J. Hu, Y. Hou, X. Gao, Sea slug inspired smart marine antifouling coating with reversible chemical bonds: Controllable UV-responsive coumarin releasing and efficient UV-healing properties, *Chem. Eng. J.* 429 (2022) 132471, <https://doi.org/10.1016/j.cej.2021.132471>.
- S. Kumari, A.K. Tyor, A. Bhatnagar, Evaluation of the antibacterial activity of skin mucus of three carp species, *Int. Aquat. Res.* 11 (3) (2019) 225–239, <https://doi.org/10.1007/s40071-019-0231-z>.
- P. Saha, A.R. Palanisamy, M. Santi, R. Ganguly, Thermoresponsive zwitterionic poly(phosphobetaine) microgels: Effect of macro-RAFT chain length and cross-linker molecular weight on their antifouling properties, *Polym. Adv. Technol.* 32 (7) (2021) 2710–2726, <https://doi.org/10.1002/pat.5214>.
- S.K. Rath, J.G. Chavan, T.K. Ghorpade, T. Umasankar Patro, M. Patri, Structure–property correlations of foul release coatings based on low hard segment content poly(dimethylsiloxane–urethane–urea), *J. Coat. Technol. Res.* 15(1) (2018) 185–198, <https://doi.org/10.1007/s11998-017-9982-x>.
- H.C. Hung, P. Jain, P. Zhang, F. Sun, A. Sinclair, T. Bai, B. Li, K. Wu, C. Tsao, E. J. Liu, H.S. Sundaram, X. Lin, A coating-free nonfouling polymeric elastomer, *Adv. Mater.* 29 (31) (2017) 1700617, <https://doi.org/10.1002/adma.201700617>.
- M. Lejars, A. Margailan, C. Bressy, Fouling release coatings: a nontoxic alternative to biocidal antifouling coatings, *Chem. Rev.* 112 (8) (2012) 4347–4390, <https://doi.org/10.1021/cr200350v>.
- Y. Xie, Z. Yang, Antitermitic and antifungal activities of eugenol and its congeners from the flower buds of *Syzygium aromaticum* (clove), *Ind. Crops Prod.* 77 (2015) 780–786, <https://doi.org/10.1016/j.indcrop.2015.09.044>.
- R. Piletti, A.M. Bugiereck, A.T. Pereira, E. Gussati, J. Dal Magro, J.M.M. Mello, F. Dalcanton, Microencapsulation of eugenol molecules by  $\beta$ -cyclodextrine as a thermal protection method of antibacterial action, *Biomater. Adv.* 75 (2017) 259–271, <https://doi.org/10.1016/j.msec.2017.02.075>.
- T. Wang, J. Zhang, Y. Cai, L. Xu, L. Yi, Protein-resistant amphiphilic copolymers containing fluorosiloxane side chains with controllable length, *ACS Appl. Polym. Mater.* 4 (10) (2022) 7903–7910, <https://doi.org/10.1021/acscapm.2c01484>.
- J. Sun, C. Liu, J. Duan, J. Liu, X. Dong, Y. Zhang, N. Wang, B. Hou, Facile fabrication of self-healing silicone-based poly(urea-thiourea)/tannic acid composite for anti-biofouling, *J. Mater. Sci. Technol.* 124 (2022) 1–13, <https://doi.org/10.1016/j.jmst.2022.01.026>.
- A.J. Birch, J.L. Stanford, The effects of hard segment content on microphase separation and physical properties of non-linear, segmented copolyureas formed by RIM, *Polym. Bull.* 22 (5) (1989) 629–635, <https://doi.org/10.1007/BF00718945>.
- W. Yan, T. Tian, M. Li, Y. Zong, Y. Liao, Depleting conjugated CC bonds in tung oil/Eucommia rubber composite film: fundamentally improve the flexibility and crack resistance for high-performance protective coating, *Ind. Crops Prod.* 191 (2023) 115943, <https://doi.org/10.1016/j.indcrop.2022.115943>.
- Y. Zhang, T. Wang, H. Si, Y. Liu, Superhydrophobic polyurea coating with excellent adhesion and abrasion resistance prepared via convenient microphase separation, *Prog. Org. Coat.* 170 (2022) 106994, <https://doi.org/10.1016/j.porgcoat.2022.106994>.
- Y. Cui, H. Pan, J. Zhang, L. Cao, C. Zong, Influence of polydimethylsiloxane on the microstructure and properties of polyester thermoplastic polyurethane, *J. Polym. Res.* 29 (6) (2022) 218, <https://doi.org/10.1007/s10965-022-03079-2>.
- A. Lucas, A. Tauleigne, F. Da Cruz-Boisson, A. Crépét, A. Bergeron-Vanhille, G. Martin, Mechanical properties enhancement while decreasing the viscosity of copolyether-ester from in situ formation of star-based structures by reactive extrusion, *Ind. Eng. Chem. Res.* 59 (38) (2020) 16579–16590, <https://doi.org/10.1021/acs.iecr.0c02801>.
- X. Jian, Y. Hu, W. Zhou, L. Xiao, Self-healing polyurethane based on disulfide bond and hydrogen bond, *Polym. Adv. Technol.* 29 (1) (2018) 463–469, <https://doi.org/10.1002/pat.4135>.

- [35] T. Nakajima, Generalization of the sacrificial bond principle for gel and elastomer toughening, *Polym. J.* 49 (6) (2017) 477–485, <https://doi.org/10.1038/pj.2017.12>.
- [36] S. Zechel, R. Geitner, M. Abend, M. Siegmann, M. Enke, N. Kuhl, M. Klein, J. Vitz, S. Gräfe, B. Dietzek, M. Schmitt, J. Popp, U.S. Schubert, M.D. Hager, Intrinsic self-healing polymers with a high E-modulus based on dynamic reversible urea bonds, *NPG Asia Mater.* 9 (8) (2017) e420–e, <https://doi.org/10.1038/am.2017.125>.
- [37] R. Chen, Q. Xie, H. Zeng, C. Ma, G. Zhang, Non-elastic glassy coating with fouling release and resistance abilities, *J. Mater. Chem. A* 8 (1) (2020) 380–387, <https://doi.org/10.1039/C9TA09794E>.
- [38] R. Guo, W. Qi, D. Li, G. Chen, Q. Li, Z. Zhou, Amino acid-modified polysiloxane for superior metal bonding and antimicrobial coatings, *Prog. Org. Coat.* 183 (2023) 107720, <https://doi.org/10.1016/j.porgcoat.2023.107720>.
- [39] W. Tian, H. Xu, Z. Guo, Polyurethane coatings modified by OH-PDMS for anti-cavitation, antifouling and anticorrosion applications, *Prog. Org. Coat.* 179 (2023) 107515, <https://doi.org/10.1016/j.porgcoat.2023.107515>.
- [40] M.G. Olivier, M. Poelman, M. Demuyne, EIS evaluation of the filiform corrosion of aluminium coated by a cataphoretic paint, *Prog. Org. Coat.* 52 (4) (2005) 263–270, <https://doi.org/10.1016/j.porgcoat.2004.05.008>.
- [41] Y. Zou, Y. Wang, S. Xu, T. Jin, D. Wei, Superhydrophobic double-layer coating for efficient heat dissipation and corrosion protection, *Chem. Eng. J.* 362 (2019) 638–649, <https://doi.org/10.1016/j.cej.2019.01.086>.
- [42] J. Sun, J. Duan, X. Liu, Environmentally benign smart self-healing silicone-based coating with dual antifouling and anti-corrosion properties, *Appl. Mater. Today*. 28 (2022) 101551, <https://doi.org/10.1016/j.apmt.2022.101551>.
- [43] Q. Mo, G. Qin, K. Ling, X. Lv, N. Wang, W. Li, Layer-by-layer self-assembled polyurea layers onto MAO surface for enhancing corrosion protection to aluminum alloy 6063, *Surf. Coat. Tech.* 405 (2021) 126653, <https://doi.org/10.1016/j.surfcoat.2020.126653>.
- [44] K. Zhu, J. Zhang, H. Zhang, H. Tan, W. Zhang, Y. Liu, H. Zhang, Q. Zhang, Fabrication of durable superhydrophobic coatings based on a novel branched fluorinated epoxy, *Chem. Eng. J.* 351 (2018) 569–578, <https://doi.org/10.1016/j.cej.2018.06.116>.
- [45] Z. Liu, C. Zhang, X. Zhang, C. Wang, F. Liu, R. Yuan, H. Wang, Durable superhydrophobic PVDF/FEVE/GO@TiO<sub>2</sub> composite coating with excellent anti-scaling and UV resistance properties, *Chem. Eng. J.* 411 (2021) 128632, <https://doi.org/10.1016/j.cej.2021.128632>.